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Archival Report

Circulating MicroRNAs in Cord Blood to Predict Attention-Deficit/Hyperactivity Disorder Diagnosis

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ABSTRACT

BACKGROUND: There are large knowledge gaps in the etiology of attention-deficit/hyperactivity disorder (ADHD), and although it is a prevalent and highly heritable neurodevelopmental disorder, diagnosis can be challenging. We aimed to assess the association of circulating blood plasma microRNAs (miRNAs) at birth with ADHD for use as biomarker candidates and build an miRNA-based prediction model.

METHODS: Our study population consisted of 206 children with ADHD (33.0% female), 207 control children (33.8% female), and their parents from the MoBa (Norwegian Mother, Father, and Child Cohort Study). Expression levels of 51 selected miRNAs in plasma from children's cord blood at birth and from both parents during early pregnancy were quantified by quantitative polymerase chain reaction and tested for association with children's ADHD diagnosis and ADHD symptom scores based on ratings by parents.

RESULTS: Seven miRNAs were differentially expressed at birth in children with ADHD and control children (false discovery rate $< .05$), and 31 had a statistically significant linear relationship with parent-rated ADHD symptom score at 8 years. A 19-miRNA ADHD prediction model achieved good discrimination in the test population (area under the receiver operating curve = 0.959, accuracy = 0.893). Functional analysis for the 19-miRNA prediction set revealed involvement in several highly relevant pathways, e.g., dopaminergic synapse, circadian rhythm, and axon guidance. We also found that parental miRNA expression levels significantly associated with children's ADHD diagnoses and/or ADHD symptoms scores.

CONCLUSIONS: We showed that expression levels of circulating miRNAs at birth may be used to predict increased risk of ADHD diagnosis, and our 19-miRNA set should be included in future efforts to develop a biomarker panel.

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Attention-deficit/hyperactivity disorder (ADHD) is a common childhood-onset neurodevelopmental disorder with a worldwide prevalence estimated at 5.9% in youth and 2.5% in adults (1). ADHD is characterized by inattention and/or hyperactivity and impulsivity at a level that interferes with typical development and functioning and is associated with, e.g., lower quality of life, difficulties socializing, academic underachievement, and unemployment. Additionally, there is an increased risk of several comorbid psychiatric diagnoses, accidental injury, and somatic illnesses such as asthma, epilepsy, and immune disorders (1,2). Meta-analyses of twin studies indicate substantial genetic involvement, with a mean heritability of 74% (3). However, ADHD is a biologically complex disorder, and studies indicate that several genes and environmental exposures each contribute to a very small increase in risk, suggesting an interplay of genetic and environmental factors, involving epigenetic mechanisms (4–6).

MicroRNAs (miRNAs) are small noncoding RNAs involved in most biological processes as regulators of gene expression at the level of translation (7). They are key players in brain development (8), and therefore there is great interest in their possible role in psychiatric or neurodevelopmental disorders, e.g., bipolar disorder, schizophrenia, major depressive disorder, and autism spectrum disorder (ASD) (9,10). Research on miRNAs in ADHD has mainly focused on biomarker identification and treatment response monitoring (11,12). Some studies have reported specific miRNAs as candidate biomarkers for ADHD (13–16).

ADHD presents itself differently across individuals, making diagnosis and treatment challenging (2,17). In early childhood, diagnosis relies heavily on how parents or other caretakers perceive the child, and in preschool children, it is difficult to separate ADHD from children's natural developmental process (18). A clinical biomarker would be a valuable tool to support the diagnosis of those difficult cases. miRNAs have been recognized as potential biomarker candidates for many different pathologies and are interesting targets for the development of an ADHD biomarker in the future (19,20). The relationship of miRNA expression level in brain versus blood is not yet understood. However, exosomes containing miRNAs

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from the brain are able to cross the blood-brain barrier, suggesting that peripheral blood miRNAs can be informative about functional states (21). Additionally, studies on psychiatric disorders have identified overlaps between miRNA expression state in brain- and blood-based samples [as reviewed in Roy *et al.* (22)], and a study of baboons found correlated patterns of miRNA expression between brain regions and peripheral blood mononuclear cells (23).

To our knowledge, there are no studies on ADHD and miRNAs using children's cord blood. One recent study on cord blood found differences in miRNA expression associated with children's cognitive and language development at 1 year of age, indicating that miRNA profiles at birth can predict outcomes of neurodevelopment (24). Early diagnosis and optimal treatment of ADHD is essential to minimize deleterious outcomes related to this disorder (1), and knowledge of molecular markers and underlying epigenetic mechanisms hold significant promise in this regard. Investigating miRNA expression levels at birth, along with parental expression levels, could reveal how they influence development of ADHD.

In the current study, we utilized a large population-based birth cohort and aimed to estimate the association of circulating miRNAs in children's cord blood plasma with ADHD and ADHD symptom pressure and to build an miRNA-based prediction model. Additionally, we aimed to estimate the association between miRNA expression profiles in parental blood plasma and their children's ADHD.

METHODS AND MATERIALS

Study Population

The MoBa (Norwegian Mother, Father, and Child Cohort Study) is a population-based pregnancy cohort study conducted by the Norwegian Institute of Public Health. Participants were recruited from all over Norway from 1999 to 2008. The women consented to participation in 41% of the pregnancies. The cohort includes approximately 114,500 children, 95,200 mothers, and 75,200 fathers. Establishment of MoBa and initial data collection was based on a license from the Norwegian Data Protection Agency and approval from the Regional Committees for Medical and Health Research Ethics. The MoBa cohort is currently regulated by the Norwegian Health Registry Act. Our study was approved by the Regional Committees for Medical and Health Research Ethics.

The current substudy of MoBa is based on 206 children with ADHD and 207 sex-, birth year–, and birthplace-matched control children and their parents. Children with ADHD and control children were matched on sex due to the tendency for boys to be diagnosed earlier than girls, on birth year due to possible changes in general public knowledge and diagnostic process in the years since MoBa started, and on birthplace to avoid certain regions only represented by either cases or controls, as MoBa participation is higher in the west and south-east. The population was selected by availability of plasma from children's cord blood samples and from parental blood samples taken during gestation weeks 17 to 18. Exclusion criteria included presence of epilepsy, cerebral palsy, or ASD recorded in questionnaires or Down syndrome or serious congenital malformations registered in the Medical Birth Registry of Norway—a national health registry containing information about all births in Norway. Information on ADHD diagnosis was reported according to ICD-10 (25) and obtained from the Norwegian Patient Registry—a mandatory national health care registry for reports on individual health care data from specialized health care services. ADHD was defined as at least 2 separate entries of the ICD-10 diagnosis hyperkinetic disorder (code F90). MoBa questionnaires included measurement of parents' civil status, education, and smoking habits. Missing values in variables from questionnaires and plasma quality variables were replaced by imputation using the R package missForest (26). Two parent-rated scales with information regarding ADHD-related traits and symptoms were also included in questionnaires: the Child Behavior Checklist (CBCL) (27) at 18 months and 3 years and the Rating Scale for Disruptive Behavior Disorders (RS-DBD) (28) at 8 years. This allowed for estimation of the association between miRNA expression level at birth and later ADHD symptom pressure. Details regarding the rating scales are provided in the Supplement.

Blood Plasma miRNA Expression

The 56 miRNAs included in this study (Table S1) were selected from previously published literature [summarized in Dypås *et al.* (29) and updated in Table S2], with a focus on 2 of our own publications (29,30). miRNAs showing overlap between different publications were prioritized, along with novel candidates only identified in our publication based on gene target prediction (30). Additionally, miRNAs of the miR-17/92 cluster were of interest due to their importance in normal fetal development and other processes relevant for ADHD (31). Selection was limited by the small amount of starting material and the availability of miRNA-specific primers in the oligo-primer bank developed in the early stages of this project. Experimental procedures, including hemolysis assessment, RNA isolation, preamplification, quantification by quantitative polymerase chain reaction, and data preprocessing and quality assessment, are described in the Supplement. A total of 5 miRNAs were removed during data quality assessment (Table S1).

Statistical Analysis

Bivariate association analysis was performed by *t* test, Spearman's rank correlation, χ^2 test, or Fisher's exact test between ADHD status, parent-rated ADHD symptom scores, and the demographic variables included in Table 1. This was also applied to test the association of miRNA expression levels with health region and degree of hemolysis measured by spectrophotometric absorbance by oxyhemoglobin at 414 nm (A414) and visual grading.

Differential miRNA expression level in participants with ADHD and control participants was first evaluated by *t* tests with multiple testing correction applied using the Benjamini-Hochberg method with a false discovery rate (FDR) $< .05$. Benjamini-Hochberg was also used for all other statistical tests requiring multiple testing correction, and the p_{FDR} is given in the results. Linear regression was performed to evaluate possible linear relationships between the parent-rated ADHD symptoms scales and miRNA expression levels. *t* Tests and linear regression were also performed to evaluate differential miRNA

Table 1. Population Demographics and Bivariate Association Analysis With ADHD Status

	Control, n = 207	ADHD, n = 206	Total, N = 413	p Value	r_s
Sex					
Female	70 (33.8%)	68 (33.0%)	138 (33.4%)	.945[a]	–
Male	137 (66.2%)	138 (67.0%)	275 (66.6%)		
Birth Weight, g					
Median [min–max]	3710 [2000–5000]	3610 [1100–5000]	3670 [1100–5000]	.137[b]	–
Gestational Age, Weeks					
Median [min–max]	40 [35–43]	40 [27–45]	40 [27–45]	.559[b]	
Missing	34 (16.4%)	32 (15.5%)	66 (16.0%)		
Health Region					
South-East	98 (47.3%)	98 (47.6%)	196 (47.5%)	.409[a]	–
West	74 (35.7%)	73 (35.4%)	147 (35.6%)		
Middle	11 (5.3%)	18 (8.7%)	29 (7.0%)		
North	24 (11.6%)	17 (8.3%)	41 (9.9%)		
Civil Status					
Married	112 (54.1%)	86 (41.7%)	198 (47.9%)	.058[c]	–
Divorced/separated	1 (0.5%)	2 (1.0%)	3 (0.7%)		
Cohabiting	90 (43.5%)	108 (52.4%)	198 (47.9%)		
Single	3 (1.4%)	7 (3.4%)	10 (2.4%)		
Missing	1 (0.5%)	3 (1.5%)	4 (1.0%)		
Paternal Education Level					
Low	13 (6.3%)	11 (5.3%)	24 (5.8%)	.007[a]	−0.17
Medium	128 (62.1%)	100 (48.3%)	228 (55.2%)		
High	39 (18.9%)	57 (27.5%)	96 (23.2%)		
Very high	17 (8.3%)	33 (15.9%)	50 (12.1%)		
Missing	9 (4.4%)	6 (2.9%)	15 (3.6%)		
Maternal Education Level					
Low	12 (5.8%)	5 (2.4%)	17 (4.1%)	.017[a]	−0.13
Medium	92 (44.7%)	74 (35.7%)	166 (40.2%)		
High	72 (35.0%)	85 (41.1%)	157 (38.0%)		
Very high	19 (9.2%)	34 (16.4%)	53 (12.8%)		
Missing	11 (5.3%)	9 (4.3%)	20 (4.8%)		
Did Father Smoke Before Pregnancy?					
No	141 (68.1%)	114 (55.3%)	255 (61.7%)	.010[a]	0.14
Yes	65 (31.4%)	91 (44.2%)	156 (37.8%)		
Missing	1 (0.5%)	1 (0.5%)	2 (0.5%)		
Maternal Smoking, Week 15 of Pregnancy					
No	152 (73.4%)	131 (63.6%)	283 (68.5%)	.003[c]	0.16
Sometimes	3 (1.4%)	11 (5.3%)	14 (3.4%)		
Daily	11 (5.3%)	25 (12.1%)	36 (8.7%)		
Missing	41 (19.8%)	39 (18.9%)	80 (19.4%)		
CBCL ADHD Index, 18 Months					
Median [min–max]	2.00 [0–7.00]	3.00 [0–8.00]	3.00 [0–8.00]	.003[b]	0.17
Missing	41 (19.8%)	54 (26.2%)	95 (23.0%)		
CBCL ADHD Index, 3 Years					
Median [min–max]	3.00 [0–10.0]	5.00 [0–12.0]	4.00 [0–12.0]	<.001[b]	0.32
Missing	67 (32.4%)	85 (41.3%)	152 (36.8%)		
RS-DBD Total ADHD					
Median [min–max]	7.00 [0–32.0]	24.5 [4.00–50.0]	13.0 [0–50.0]	<.001[b]	0.75
Missing	116 (56.0%)	122 (59.2%)	238 (57.6%)		

Table 1. Continued

	Control, n = 207	ADHD, n = 206	Total, N = 413	p Value	r_s
RS-DBD Inattention					
Median [min–max]	4.00 [0–15.0]	14.0 [4.00–27.0]	7.00 [0–27.0]	<.001[b]	0.75
Missing	116 (56.0%)	122 (59.2%)	238 (57.6%)		
RS-DBD Hyperactivity/Impulsivity					
Median [min–max]	2.00 [0–23.0]	11.0 [0–25.0]	6.00 [0–25.0]	<.001[b]	0.62
Missing	116 (56.0%)	122 (59.2%)	238 (57.6%)		

Values presented are n (%) or median [min–max]. Health region is defined by the division of Norway into 4 regions for health services. Parental education levels are defined as follows: low = 9-year elementary education, medium = further education of 1 to 3 years (not university), high = university/college for 4 years or less, and very high = university/college for more than 4 years. r_s indicates Spearman's rank correlation coefficient.

ADHD, attention-deficit/hyperactivity disorder; CBCL, Child Behavior Checklist; max, maximum; min, minimum; RS-DBD, Rating Scale for Disruptive Behavior Disorders.

[a]p Value from χ^2test.

[b]p Value from t test.

[c]p Value from Fisher's exact test.

expression in parents of children with ADHD versus control children and its association with child ADHD symptom scores.

Prediction models were built separately using child, maternal, or paternal miRNA expression data; 80% of the study population was used in model training, while 20% was retained as a test set. Further details regarding model choice and building and sensitivity testing are provided in the Supplement.

Gene Target and Functional Enrichment Analysis

Genes targeted by the 19 miRNAs of the final prediction model were identified using multiMiR, an miRNA-target interaction R package and database compiled from 14 different databases, including the 3 validated miRNA-target databases: miRTarBase, TarBase, and miRecords (32). We included only validated targets. The R package miRmapper was then used for clustering analysis of the miRNAs based on gene target similarity. Details can be found in the package description (33). Functional enrichment analysis of the 19-miRNA set was performed with DIANA mirPath version 4.0 using validated targets from TarBase version 8.0 (34). Analysis was run as genes union with significance set to $p_{FDR} < .05$.

RESULTS

Demographic Characteristics and Questionnaire Data

The study population and its characteristics are presented in Table 1 along with results from the bivariate association analysis. All parent-rated ADHD symptom scales were significantly correlated with ADHD diagnosis, but only the RS-DBD had a correlation coefficient indicating a strong relationship. Thus, RS-DBD was used in the linear regression analysis.

Plasma Quality Based on Evaluation of Hemolysis

Hemolysis was significantly higher for children's cord blood plasma samples compared with parental samples assessed by both A414 and visual grading (Figure S3). This was expected due to parental plasma samples being processed at the location of the blood draw, while cord blood was transported to the Norwegian Institute of Public Health prior to plasma extraction. Paternal samples had a small, but statistically significant, elevation in hemolysis assessed by A414 compared with maternal samples (1-way analysis of variance [ANOVA] followed by Tukey post hoc test, $p < .05$). Blood draws from men were found to have increased risk of hemolysis (35). Spearman's rank correlations for the 2 measures of hemolysis with miRNA expression revealed that the highest correlation was for A414 and with the highest number of miRNAs (Table S3). Several of the miRNAs correlating with A414 also correlated at a significant level ($p_{FDR} < .05$) with health region, consistent with the length of transport affecting sample quality. In a separate analysis of children's cord blood samples only, the number of miRNAs significantly correlated with both health region and A414 increased, while the number decreased for visual grading (Table S3). Based on this, A414 was chosen as best suited to adjust for the effect of hemolysis on miRNA quantification.

miRNA Expression in Children's Cord Blood Plasma Associated With ADHD

Seven miRNAs were differentially expressed at a significant level ($p_{FDR} < .05$) in participants with ADHD and control participants as evaluated by t tests (Figure 1A–G). The miRNAs hsa-let-7d-5p, hsa-miR-126-3p, hsa-miR-138-5p, hsa-miR-15a-5p, and hsa-miR-16-5p were downregulated in ADHD, while hsa-miR-34c-3p and hsa-miR-519d-3p were upregulated. Through linear regression, we identified 31 miRNAs as having a significant ($p_{FDR} < .05$) linear relationship with the RS-DBD total ADHD symptom score at 8 years (Table S4). Results were similar for RS-DBD inattention and hyperactivity/impulsivity scores, with 37 and 25 miRNAs reaching significance, respectively (Tables S5 and S6).

Feature selection resulted in 19 miRNAs for model building, including all miRNAs identified by t tests except for hsa-miR-519d-3p. Cross-validation analysis of several models (Figure S4A–C) identified random forest as most promising. After hyperparameter tuning and training the model on the training set, we achieved a model with an area under the receiver operating curve (AUC) of 0.959 and accuracy of 0.893 (sensitivity = 0.881, specificity = 0.905) for the test set (Figure 2). The 3 topmost results from linear regression with ADHD symptoms, rated by p value, were also among the most

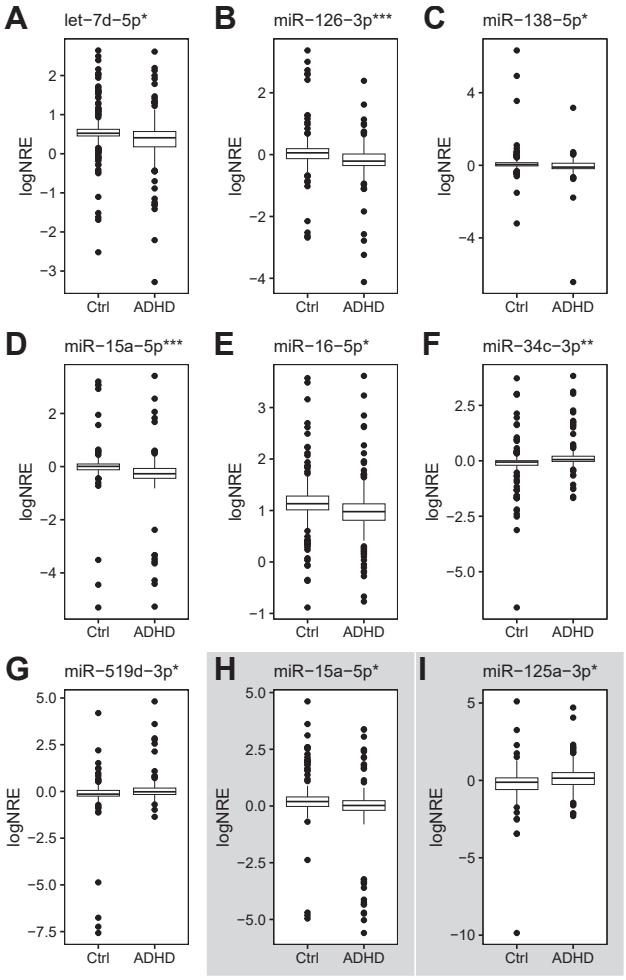


Figure 1. Differentially expressed miRNAs. Results for parental miRNA expression are shaded in gray. miRNAs significantly differentially expressed in **(A–G)** cord blood plasma from children with ADHD or Ctrl children and in blood plasma from **(H)** mothers and **(I)** fathers of children with ADHD or Ctrl children. Evaluated by *t* tests with significance set to $p_{FDR} < .05$. *$p_{FDR} < .05$, **$p_{FDR} < .01$, ***$p_{FDR} < .001$. ADHD, attention-deficit/hyperactivity disorder; Ctrl, control; FDR, false discovery rate; logNRE, log_2-transformed normalized relative expression; miRNA, microRNA.

important features of the prediction model (Figure 2B), and their linear relationships are presented in Figure 3.

Multivariate ANOVA was used in sensitivity testing, and neither sex, gestational age, birth weight, parental education level, parental smoking habits, or health region were identified as modifiers of the association between miRNA expression and ADHD at a significant level ($p > .05$). To check whether there was a large discrepancy in misclassification by sex, the final prediction model was tested separately on female and male participants, with an accuracy of 0.893 and 0.839, respectively. The addition of gestational age, birth weight, parental education level, parental smoking habits, or health region as covariates in linear regression analysis for predicting RS-DBD ADHD symptom scores had little to no effect on most results and did not indicate strong positive confounding (Tables S7–S13). Of the 3 topmost miRNAs, the only instance with considerable change in slope (>10%) was observed for hsa-miR-15a-5p with inclusion of maternal smoking in the model. However, this change was more consistent with suppression or negative confounding, which was true overall for most instances of >10% change in slope. Tables S14 to S16 also present the effect of including sex in the main models.

miRNA Expression in Parental Blood Plasma Associated With Children's ADHD

miRNAs in blood plasma from parents were differentially expressed at a significant level for hsa-miR-15a-5p downregulated in mothers and hsa-miR-125a-3p upregulated in fathers of children with ADHD (Figure 1H, I). Linear regression identified 22 maternal miRNAs and 22 paternal miRNAs as having a significant linear relationship with children's RS-DBD total ADHD symptom scores at 8 years (Tables S17 and S18). Feature selection identified 13 and 19 miRNAs for model building with expression levels in mothers and fathers, respectively. Cross-validation of all models built using maternal or paternal miRNA expression revealed low accuracy for all models, but the pattern of performance was like that of children's miRNA expression. Random forest was chosen as the final model (Figure S4D–I). We achieved AUCs of 0.805 and 0.818 and accuracies of 0.707 and 0.679 for maternal and paternal models, respectively (Figure S5).

Analysis of Biological Relevance

Genes targeted by our 19-miRNA set were identified. The clustering dendrogram based on gene target similarity shows miRNA families clustered together (Figure 4); 2 members of the mir-15/107 family, hsa-miR-15a-5p and hsa-miR-16-5p, and 2 members of the let-7 family, hsa-let-7b-5p and hsa-let-7d-5p. Another group was the miR-17/92 cluster, a group of physically adjacent miRNAs transcribed together, including hsa-miR-17-5p, hsa-miR-20a-5p, and hsa-miR-92a-3p, and hsa-miR-20b-5p in a paralog cluster.

Functional enrichment analysis of the gene targets revealed involvement in highly relevant pathways from the Kyoto Encyclopedia of Genes and Genomes (36), such as dopaminergic synapse, axon guidance, circadian rhythm, circadian entrainment, Wnt- and mTor (mechanistic target of rapamycin) signaling, long-term depression, alcoholism, and the neurotrophin signaling pathway (Table S19).

DISCUSSION

In this study, we identified several miRNAs in children's cord blood plasma associated with ADHD diagnosis later in life and built a 19-miRNA ADHD prediction model with relatively high predictive ability. miRNA expression profiles in parental blood plasma from weeks 17 to 18 of pregnancy were also shown to be associated with children's ADHD. Additionally, we found the expression levels of 31 miRNAs at birth to have a linear relationship with ADHD symptom scores based on ratings by parents when the children were age 8 years. To the best of our knowledge, this is the first study to investigate whether miRNA expression status at birth can be shown to be associated with an ADHD diagnosis, as well as incorporating parental miRNA levels in early pregnancy.

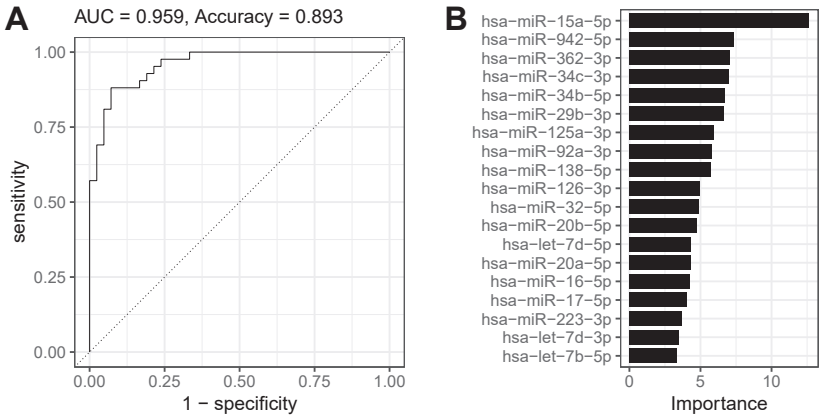


Figure 2. Attention-deficit/hyperactivity disorder prediction model built using expression levels of 19 miRNAs in children's cord blood plasma. **(A)** Receiver operating characteristic curve for the model built using the machine learning algorithm random forest. **(B)** Feature importance ranked by mean decrease in Gini impurity. AUC, area under the receiver operating curve; miRNA, microRNA.

Previously identified miRNAs associated with ADHD have been investigated in children, adolescents, and adults, and therefore, we included several miRNAs that have only been associated with ADHD through ADHD-related genes and target prediction (30). The significance of these miRNAs in our current study could mean that they are important during development of ADHD as opposed to the continued expression of ADHD symptomology. We consider this a unique strength of our study. We performed multiple testing correction in all statistical tests to further strengthen findings. Extensive sensitivity testing was also included to map the effect of several relevant variables. Hemolysis is an important preanalytical variable, as lysis of red blood cells and leakage of cell contents into the surrounding plasma prior to sample processing is known to substantially skew miRNA content in the otherwise cell-free plasma (37–39). We included 2 methods of hemolysis assessment to enable correction for this effect. ADHD is highly comorbid with other neurodevelopmental or psychiatric disorders, especially ASD (40). We excluded children with ASD to remove noise from comorbidity, and future efforts with larger study populations need to include comorbidities and investigate the specificity of our biomarker panel.

Our 19-miRNA model was able to predict, with relatively high accuracy, which children would receive an ADHD diagnosis later in life based on miRNA status at birth. This is a promising result toward achieving earlier and easier diagnosis of ADHD. Wang *et al.* (12,13) have previously developed ADHD classification models based on miRNA expression profiles in white blood cells of children ages 6 to 12 years. Two miRNAs in our model, hsa-miR-92a-3p and hsa-miR-223-3p, were also in one or both of their models, respectively, and should be included in future efforts to develop an age-independent biomarker panel. Additionally, in their models, hsa-miR-126-5p originates from the same pre-miRNA as hsa-miR-126-3p of our 19-miRNA set. In concordance with their results of hsa-miR-126-5p being upregulated in ADHD, we found hsa-miR-126-3p to be positively correlated with increasing symptom pressure in linear regression with ADHD symptoms at 8 years. Furthermore, Zhu *et al.* (16) developed a prediction model using a different set of miRNAs. Their 5-miRNA profile had high diagnostic accuracy, and follow-up after 3 to 6 months of methylphenidate treatment showed that expression levels had moved closer to that of neurotypical control children. Participants were also separated by ADHD presentation: combined, predominantly hyperactive, and predominantly inattentive. The model still performed well but with lower validity for the latter two, highlighting the importance of ADHD presentation. Gene target analysis for their 5-miRNA set resulted in 612 unique genes, and 583 of them overlap with genes targeted by our 19-miRNA set, with 13 of our miRNAs targeting 100 to 500 of these genes (Table S20). Thus, the different models potentially share a substantial degree of regulatory function. Combined with these studies, we have demonstrated the feasibility of an miRNA signature for

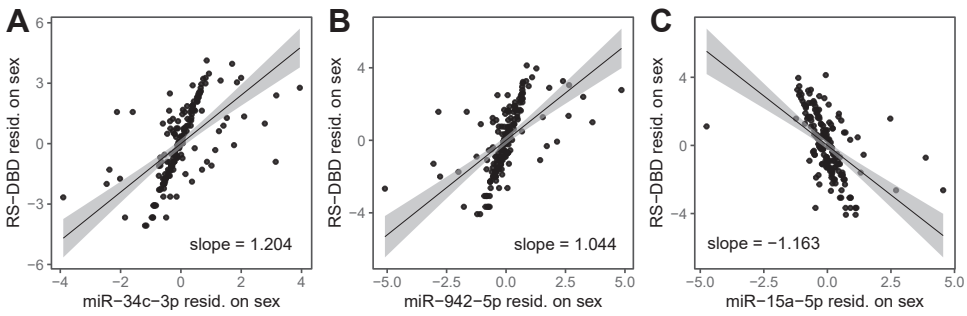


Figure 3. Linear relationship of miRNA expression level with RS-DBD ADHD symptom scores adjusted for sex. **(A–C)** The top 3 significant miRNAs are presented. Evaluated by linear regression with significance set to false discovery rate $<$ .05 and sex included as covariate. Both miRNA expression (log_2-transformed normalized relative expression) in children's cord blood plasma and the RS-DBD ADHD symptom score are residualized on the covariate sex to visualize the full model. ADHD, attention-deficit/hyperactivity disorder; miRNA, microRNA; RS-DBD, Rating Scale for Disruptive Behavior Disorders.

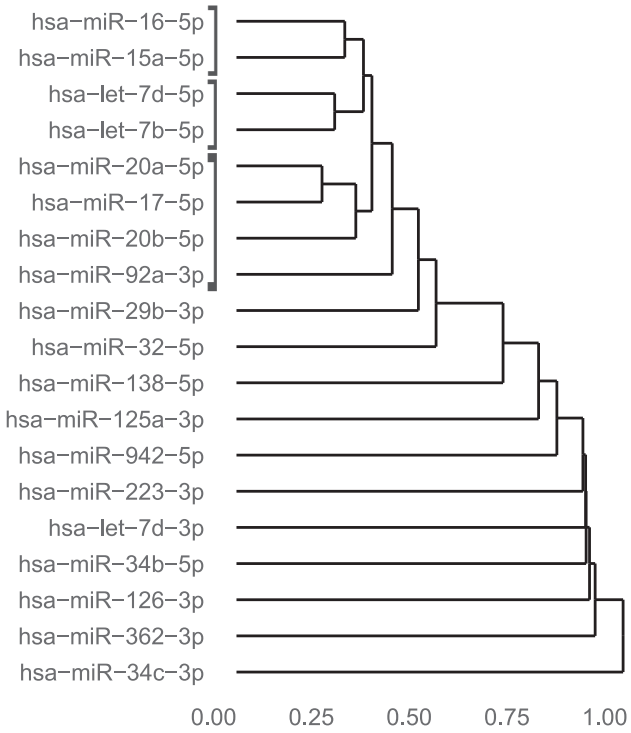


Figure 4. Clustering dendrogram based on gene target similarity for our 19-miRNA set. Distance is based on the similarity of the miRNAs' Jaccard index values to each other and was calculated using the R package miRmapper (33). miRNA families and clusters mentioned in the text are indicated by brackets. miRNA, microRNA.

ADHD, while the challenge lies in finding the right miRNAs across age, sex, ethnicity, comorbidities, etc.

In our previous study of children (mean age 8.2 years) across 6 European birth cohorts, we identified 29 miRNAs in whole blood associated with parent-rated hyperactivity on the Conners Rating Scale (29), with 14 of those included here. Linear regression revealed that 5 of them were associated with ADHD symptom scores, consistent with previous findings; hsa-miR-19a-3p, hsa-miR-19b-3p, hsa-miR-24-3p, hsa-miR-29a-3p, and hsa-miR-29c-3p decreased with increasing symptoms scores. However, none of these were included in our 19-miRNA prediction model. Four other miRNAs, hsa-miR-142-3p, hsa-miR-29b-3p, hsa-miR-92a-3p, and hsa-miR-942-5p, were positively associated with ADHD symptoms, the opposite direction of the association with Conners hyperactivity scores in the previous study, although they were among the 19-miRNA set. This could be interpreted as the first 5 miRNAs possibly being age-independent markers of ADHD symptom pressure, while the other 4 have different expression profile or functionality at birth compared with later in childhood.

The miRNAs hsa-let-7d-5p, hsa-miR-126-3p, hsa-miR-138-5p, hsa-miR-15a-5p, and hsa-miR-16-5p were downregulated in cord blood plasma of children diagnosed with ADHD, while hsa-miR-34c-3p and hsa-miR-519d-3p were upregulated. Consistent with our findings, hsa-miR-138-1, one of two pre-miRNAs from which hsa-miR-138-5p is derived, was downregulated in blood plasma samples of Egyptian children with ADHD (41); miR-138-5p was downregulated in the prefrontal cortex of spontaneously hypertensive rats, a rat model of ADHD (42); and hsa-let-7d was downregulated in whole blood of Turkish children with ADHD (14). These miRNAs have also been found to be upregulated in serum samples of children and adolescents with ADHD (11,43,44). Additionally, hsa-miR-34c-3p has been identified as upregulated in ADHD in peripheral blood mononuclear cells but downregulated in plasma (41,45). Comparison across studies is complicated due to factors such as different sample types and mostly small and demographically/geographically homogeneous samples, and there are still only a handful publications on the subject. Most importantly, we are investigating expression status at birth and comparing it with results from samples taken much later in development. The remaining miRNAs significantly differentially expressed, hsa-miR-15a-5p, hsa-miR-16-5p, and hsa-miR-519d-3p, have only been associated with ADHD in our earlier study based on gene target prediction (30). They could be important for ADHD in early neurodevelopment, as they have not appeared in studies on children or adults. The miRNAs differentially expressed in mothers or fathers of children with ADHD, hsa-miR-15a-5p and hsa-miR-125a-3p, respectively, have also been identified through target prediction only. These could be significant through affecting development of ADHD in the child, or, as ADHD is highly heritable, they could be strong biomarkers of ADHD persisting into adulthood.

The high number of miRNAs with a significant linear relationship with ADHD symptom scores, compared with only 7 significantly differentially expressed when comparing means, may point to an important feature of ADHD. An ADHD diagnosis is based on a range of traits and symptoms and the combined severity of each one (46). These are traits and symptoms that exist in the general population, and an ADHD diagnosis is only reached when they reach a certain threshold of severity. This dimensional view of ADHD as an extreme of continuous heritable traits is also supported by a large genome-wide association study meta-analysis in which ADHD risk genes overlapped with ADHD symptoms and risk behaviors in the population (4). Therefore, some participants with ADHD may be relatively close to neurotypical participants on this scale of varying severity for several different traits.

Investigating circulating miRNAs at birth was also aimed at revealing miRNAs important during early neurodevelopment and thus ADHD development. Several miRNAs of our 19-miRNA set are members of the miR-17/92 cluster, a genomic cluster essential for fetal development and is implicated in neuronal plasticity and axonal outgrowth and guidance (31). Clustered miRNAs may contribute toward physiological effects in amplified strength through coordinated coexpression and thus coregulation of several gene targets. Functional enrichment analysis has revealed involvement in several highly relevant pathways, such as dopaminergic synapses, axon guidance, and the Wnt- and mTor-signaling pathways. All have been implicated as contributors to ADHD etiology or through the action of ADHD medication (47–50). Circadian rhythm and circadian entrainment are other highly relevant pathways. There is high comorbidity of sleep disorders with ADHD, and genes related to circadian function have been implicated in ADHD (51,52).

Long-term depression and alcoholism are psychiatric disorders known to overlap with ADHD in both genetic risk and co-occurrence (53,54). Several instances of biological relevance further indicate possible prominent roles for these miRNAs in ADHD etiology.

This study has limitations that should be considered. First, although our study population includes participants from across all 4 health regions in Norway, MoBa is affected by selection bias due to attrition, and some demographic variables differ from that of the total general population (55). Two studies have found that despite this, the between-variable bivariate associations were not biased and that studies on ADHD in MoBa may have reasonable generalizability to the general child population (56,57). Nonetheless, we cannot say whether our results can be generalized across ethnicities and countries. The prediction model may also perform differently in a population reflective of an ADHD prevalence of approximately 5%. A model for diagnostic use will need to be trained and tested on such populations. We did not adjust for possible confounding by parental smoking habits and education level, as their location on the causal pathway is not known (58). However, sensitivity testing did not indicate a significant effect on the most central findings. Lastly, the ICD-10 diagnosis code F90 has no information on ADHD presentation. Splitting cases into subgroups depending on type of ADHD presentation has been shown to increase separation of ADHD cases from neurotypical controls in neuropsychological tests (59). Genetic associations with ADHD presentation also indicate actual biological difference (60).

One last interesting topic to consider is the link between placental integrity and cord blood miRNAs. Placental miRNA expression levels and pathology have been linked to a shift in cord blood miRNA profile (61,62). Placental pathology and miRNA profiles in placenta have also been found to be associated with ASD diagnosis (63,64). With the high rate of comorbidity between ADHD and ASD, this potential connection is highly relevant for future studies on causes underlying the differences observed here in cord blood plasma miRNA profiles of children with ADHD and control children.

Conclusions

Expression levels of circulating miRNAs at birth are significantly associated with, and may be able to predict, later ADHD diagnosis. The 19-miRNA set proposed here are candidates for differentiation between children with ADHD and neurotypical children. We found a linear relationship between miRNA expression at birth and parent-rated ADHD symptoms at 8 years. Parental expression levels of circulating miRNAs measured during early pregnancy were also found to be associated with children's ADHD. The results strongly suggest that miRNAs should be included in future studies on the biological mechanisms underlying ADHD. Identification of miRNAs in ADHD and understanding their role in the development or maintenance of ADHD could play an important role in both easier diagnosis and new treatment options.

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LBD processed and interpreted all data, performed statistical analyses, created visualizations, and drafted the initial article. ND contributed as project administrator. All authors contributed to interpretation of results, critically reviewed the article, and approved the final version of the article.

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